Physiology-Guided Temporal Fusion for Cross-Subject EEG–fNIRS Classification

Antigravity BCI Research Collaborative

***Abstract*—Simultaneous electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS) capture complementary electrophysiological and hemodynamic views of human cortical activation. However, disparate temporal dynamics—millisecond electrical transients versus multi-second vascular delays—impair cross-subject decoding when unconstrained attention overfits individual timing alignments. We present NVC-DelayFormer, an end-to-end temporal fusion network parameterized by biophysical neurovascular coupling (NVC) priors for subject-independent BCI decoding. The architecture pairs an authentic 2D-EEGNet spatio-temporal backbone with a continuous fNIRS Conv1D encoder into a multi-head cross-attention mechanism governed by dispersive hemodynamic delay priors ($\tau_h \in [3.5, 6.5]$ s) and strict physical causal masks ($t_{\text{nirs}} \geq t_{\text{eeg}}$). An uncertainty-aware entropy-guided gating network dynamically arbitrates between deep cross-modal tokens and shallow biophysical spectral summaries without heuristic hyperparameter tuning. Evaluated across 29 participants on the TU Berlin mental arithmetic benchmark (1,740 trials) under leave-one-subject-out (LOSO) cross-validation, NVC-DelayFormer achieved a primary 3-seed grand mean accuracy of 78.22% ± 1.00% (95% CI: [75.42, 81.02]%, Macro F1: 77.99%; primary single seed (Seed 42): 79.37% ± 7.73%). This performance significantly outperformed authentic 2D-EEGNet (72.89% ± 3.49%, $p = 1.10 \times 10^{-4}$), covariance-engineered Shrinkage-LDA with Euclidean alignment (72.82% ± 11.61%, $p = 0.0167$), shared-encoder feature concatenation (72.84% ± 0.53%, $p = 6.35 \times 10^{-4}$), and unconstrained cross-attention (72.49% ± 1.10%, $p = 7.18 \times 10^{-4}$). Across two external evaluation paradigms comprising 3,612 held-out trials under identical participant-isolated LOSO protocols, the full architecture demonstrated robust generalization: 73.77% ± 15.44% on n-back Working Memory (significantly outperforming authentic 2D-EEGNet at 68.27% by +5.50 pp, $p = 0.0357$, and Shrinkage-LDA at 62.13% by +11.64 pp, $p = 0.0022$) and 66.32% ± 12.18% on Motor Imagery (outperforming Shrinkage-LDA at 58.05% by +8.27 pp, $p = 0.0014$; Multimodal SVM at 55.63% by +10.69 pp, $p < 0.0001$; and 2D-EEGNet at 62.18% by +4.14 pp, $p = 0.0163$). Parameterized with 63,940 weights and 24.84 M MACs, forward execution requires 2.22 ms median GPU latency, establishing an accurate, calibration-free temporal fusion framework for hybrid brain–computer interfaces, facilitating future translation to bedside assistive neurotechnologies.**

***Index Terms*—Brain–computer interface, neurovascular coupling, cross-subject decoding, electroencephalography, functional near-infrared spectroscopy, temporal fusion.**

I. Introduction

CROSS-SUBJECT cognitive decoding evaluates whether neural representations learned from a source cohort reliably generalize to unseen individuals without subject-specific calibration. In hybrid brain–computer interfaces (BCIs), combining electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS) captures both fast postsynaptic electrical transients and localized capillary oxygenation dynamics [1]–[4]. However, these modalities operate across fundamentally mismatched physical timescales: neuronal depolarization elicits spectral shifts within tens of milliseconds [5], [6], whereas neurovascular coupling (NVC) and capillary perfusion peak over multi-second hemodynamic delays [7]–[9]. In calibration-free, leave-one-subject-out (LOSO) training regimes, unconstrained neural networks exploit spurious temporal alignments, impairing generalization across held-out participants.

Conventional multimodal fusion relies on manual feature engineering, concatenating bandpower summaries and temporal slope statistics into linear discriminant analysis or support vector machines [1], [10], [11]. While Riemannian geometry and Euclidean alignment (EA) mitigate inter-subject covariance shifts [12]–[15], handcrafted pipelines treat temporal windows statically, failing to resolve the dynamic latency linking neuronal excitation to delayed capillary hyperemia.

Deep learning architectures capture complex spatio-temporal representations directly from raw biosignals [16]–[18]. In electrophysiology, compact convolutional architectures such as EEGNet [16], attention-based networks such as ATCNet [19], transformer backbones like EEG Conformer [20], and modern convolutional variants like EEGConvNeXt [21] extract spatial filters across scalp montages. For optical signals, continuous-wave convolutional networks separate oxygenated (HbO) and deoxygenated (HbR) hemoglobin dynamics [22], [23]. When fusing both streams, standard cross-attention mechanisms [24], [25] compute pairwise token interactions without inductive constraints. However, unconstrained cross-attention permits unphysiological alignments, including acausal queries where future hemodynamic states attend to subsequent electrophysiology, producing optimization artifacts that fail to transfer across subjects.

Biophysical models of neurovascular coupling, established by Buxton's Balloon model [7] and Glover's deconvolution kinetics [26], demonstrate that capillary dilation and local blood oxygenation lag neuronal firing by approximately 3.5 to 6.5 seconds. Incorporating continuous Gaussian delay priors and strict physical causal masks ($t_{\text{nirs}} \geq t_{\text{eeg}}$) restricts attention to biologically plausible receptive windows. In parallel, dual-stream architectures that pair deep attentive representations with explicit sub-band oscillatory energy and optical slope summaries ground network decisions in biophysical baselines, stabilizing training against inter-subject variance.

This study presents NVC-DelayFormer, a physiology-guided

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temporal fusion network for calibration-free cross-subject decoding. The contributions are:

1) **Biophysically constrained causal cross-attention:** Pairing an authentic 2D-EEGNet backbone with a continuous optical encoder, we formulate multi-head cross-attention governed by continuous Gaussian hemodynamic delay priors ($\tau_h \in [3.5, 6.5]$ s) and strict physical causal masks ($t_{\text{nirs}} \geq t_{\text{eeg}}$), enforcing physical temporal precedence in cross-modal alignment.
2) **Uncertainty-aware entropy-guided gating:** A learned bottleneck gating network arbitrates between deep cross-modal tokens and shallow biophysical spectral summaries based on predictive distribution entropy, eliminating heuristic scaling hyperparameters.
3) **Comprehensive multi-seed and multi-paradigm validation:** Evaluated across 29 participants on the TU Berlin benchmark under LOSO cross-validation, the model achieves a primary 3-seed grand mean accuracy of 78.22% ± 1.00% (primary single seed (Seed 42): 79.37% ± 7.73%), significantly exceeding authentic 2D-EEGNet (72.89% ± 3.49%, $p = 1.10 \times 10^{-4}$), Shrinkage-LDA with EA (72.82%, $p = 0.0167$), shared-encoder feature concatenation (72.84% ± 0.53%, $p = 6.35 \times 10^{-4}$), and unconstrained cross-attention (72.49% ± 1.10%, $p = 7.18 \times 10^{-4}$). Across two independent external cohorts encompassing 3,612 held-out trials under LOSO protocols, the full architecture attains 73.77% ± 15.44% on n-back Working Memory and 66.32% ± 12.18% on Motor Imagery, establishing consistent multi-paradigm transfer with 2.22 ms median GPU latency.

II. Related Work

A. Multimodal EEG–fNIRS Neuroimaging

Early hybrid EEG–fNIRS brain–computer interfaces demonstrated that combining electrophysiological and hemodynamic recordings improves single-trial classification over either modality alone [1], [10]. Open-access benchmarks established by Shin et al. [2], [27] accelerated multimodal algorithm development by providing synchronized 30-channel EEG and 36-channel dual-wavelength optical recordings across mental arithmetic, motor imagery, and working memory tasks. Open-source instrumentation advances by von Lühmann et al. [28] and clinical fNIRS reviews by Pinti et al. [29] further confirmed the viability of wearable multimodal caps for bedside and neurorehabilitation deployment.

B. Deep Learning for Biosignal Decoding

Deep neural networks have progressively superseded handcrafted spatial filtering in electrophysiological decoding [18]. Lawhern et al. introduced EEGNet [16], employing depthwise and separable convolutions to extract frequency-band and spatial-topographic filters with minimal parameter overhead. Recent architectures incorporate attention mechanisms to capture longer temporal dependencies: Altaheri et al. developed ATCNet [19], combining temporal convolutional networks with multi-head self-attention; Song et al. proposed EEG Conformer [20], fusing local convolutional features with global self-attention; and Acharya et al. introduced EEGConvNeXt [21] for automated neurodegenerative biomarker identification. Salim and Tiwari [30] demonstrated lightweight convolution-transformers for cross-patient seizure detection, while Kostas et al. [31] and Banville et al. [32] explored self-supervised representation learning from large-scale EEG archives. For optical imaging, Pandey et al. designed fNIRSNET [23], separating oxy- and deoxyhemoglobin representations through multi-view convolutional pathways. In multimodal fusion, early deep architectures applied feature concatenation across parallel CNN branches [33], [34]. Recent works introduced dedicated cross-modal interaction mechanisms: Kwak et al. designed FGANet [35], using fNIRS temporal representations to guide electrophysiological attention; Sun et al. proposed EFMLNet [36], enforcing cross-modal mutual information learning; Li et al. developed MBC-ATT [37], implementing multi-branch convolutional pathways with bidirectional attentional fusion; and Wang et al. [25] explored symmetric cross-modal transformers. Explainable attribution methods, including Shapley-value analyses by Pérez-Velasco et al. [38], have confirmed that attention weights reveal functional neural interactions.

C. Biophysical Modeling of Neurovascular Coupling

Neuronal activation increases local metabolic demand, triggering active arteriolar vasodilation and increased cerebral blood flow that washes out paramagnetic deoxyhemoglobin while elevating oxyhemoglobin [8], [9], [39]. Buxton's Balloon model [7] formalizes this coupling as a dynamic compliance system where blood volume and venous outflow lag vasodilatory stimulation. Glover [26] and Friston et al. [40] demonstrated that the impulse hemodynamic response function (HRF) follows a double-gamma distribution with modal response peak at 3.5–6.5 s followed by post-stimulus undershoot. When stimulated by sustained task blocks (e.g., 10.0 s task engagement), the observed hemodynamic profile represents the temporal convolution of a boxcar function with the impulse HRF, reaching an extended hyperoxygenation plateau across 6.0–12.0 s [22], [41]. Grounding cross-attention in these physical latency regimes prevents acausal overfitting in data-limited human BCI studies.

III. Methods

NVC-DelayFormer maps paired multichannel EEG and fNIRS trials into cognitive states through a biophysically parameterized temporal fusion network. In this formulation, bold uppercase symbols denote multi-dimensional tensors, bold lowercase symbols denote vectors, and italic symbols denote scalars. The mini-batch dimension is denoted by B.

A. Input Data and Preprocessing

The primary benchmark uses the TU Berlin dataset [2], recording 29 healthy participants performing mental arithmetic (MA) and resting state across 1,740 trials (30 trials per condition). The scalp montage retains 30 EEG channels under the 10–5 international system, excluding ocular channels.

Continuous EEG signals are bandpass-filtered between 0.5 and 45 Hz using a zero-phase fourth-order Butterworth filter in

second-order sections, followed by common average referencing (CAR). Ten-second epochs extracted from task onset are Fourier-resampled to 1,000 samples at 100 Hz ($f_e = 100$ Hz) and baseline-corrected by subtracting the preceding 2.0 s pre-stimulus mean (-2.0 to 0.0 s).

Optical data record raw intensities across 36 channels at 760 and 850 nm, bounded below by 10^{-4} prior to optical-density conversion ($\Delta\mathrm{OD} = -\ln(I/I_0)$ relative to the session mean intensity I_0 [2]). In online streaming mode, reference intensity I_0 is evaluated from the pre-trial resting baseline $[-2.0, 0.0]$ s, eliminating session lookahead. Optical-density traces are filtered between 0.01 and 0.10 Hz. Twelve-second optical epochs extracted from task onset (incorporating a 2.0 s hemodynamic latency tail window to capture post-stimulus capillary transit) are converted into oxyhemoglobin ($\Delta[\mathrm{HbO}]$) and deoxyhemoglobin ($\Delta[\mathrm{HbR}]$) concentrations via the modified Beer–Lambert law (MBLL) [42]. The resulting 72-channel representation is Fourier-resampled to 124 samples ($f_n = 124/12$ Hz), yielding standardized input tensors $\mathbf{X}_e \in \mathbb{R}^{B\times30\times1000}$ and $\mathbf{X}_n \in \mathbb{R}^{B\times72\times124}$. In sequence truncation evaluations where the post-task tail window is omitted, optical sequences are strictly truncated to 10.0 s (104 samples at 10.33 Hz), corresponding to the 78.74% baseline.

B. Optical Conversion and Numerical Convention

The modified Beer–Lambert law relates differential optical density to hemoglobin concentrations through differential pathlength factors (DPF) [42]:

$$\mathbf{c} = \mathbf{R}\mathbf{o}, \qquad \mathbf{R} = (d_{\mathrm{num}}\mathbf{D}_{\mathrm{DPF}}\mathbf{E}_{\mathrm{num}})^{-1}, \tag{1}$$

where $\mathbf{o} = [\Delta\mathrm{OD}_{760}, \Delta\mathrm{OD}_{850}]^\top$, $\mathbf{c} = [\Delta c_{\mathrm{HbO}}, \Delta c_{\mathrm{HbR}}]^\top$, $d_{\mathrm{num}} = 30$ mm, and empirical coefficients are:

$$\mathbf{E}_{\mathrm{num}} = \begin{bmatrix} 1.4866 & 3.8437 \\ 2.5264 & 1.7989 \end{bmatrix}, \quad \mathbf{D}_{\mathrm{DPF}} = \mathrm{diag}(6.51, 5.86). \tag{2}$$

Evaluating $\mathbf{R}$ across all 36 optode pairs produces 36 HbO and 36 HbR continuous traces in standardized millimolar units.

C. Multimodal Spatio-Temporal Convolutional Encoding

Electrophysiological inputs $\mathbf{X}_e \in \mathbb{R}^{B\times30\times1000}$ are encoded via an authentic 2D-EEGNet backbone [16]. A temporal convolution (kernel 1×64, $F_1 = 8$) extracts frequency-band dynamics, followed by batch normalization and a depthwise spatial convolution (kernel $C_e \times 1$, depth factor $D = 2$, 16 spatial filters) that isolates scalp topography. After ELU activation, average pooling (kernel 1×4), and spatial dropout ($p = 0.25$), a separable convolution (depthwise 1×15 with symmetric padding 7, pointwise 1×1) projects representations to 64 dimensions, followed by batch normalization, ELU, and dropout, yielding token representations:

$$\mathbf{H}_e \in \mathbb{R}^{B\times250\times64}, \qquad f_e^{\mathrm{eff}} = 25 \text{ Hz}. \tag{3}$$

The optical encoder applies a spatial Conv1D ($72 \to 32$), batch normalization, ELU, and dropout, followed by a temporal Conv1D (kernel 5, padding 2, $32 \to 64$), batch normalization, and ELU, producing:

$$\mathbf{H}_n \in \mathbb{R}^{B\times124\times64}, \qquad f_n = 124/12 \text{ Hz}. \tag{4}$$

Because both sequences synchronize to task onset ($t = 0.0$ s), temporal indices correspond to physical timestamps:

$$t_i^n = i/f_n, \quad t_j^e = j/f_e^{\mathrm{eff}}, \quad \delta_{ij} = t_i^n - t_j^e. \tag{5}$$

D. Physiology-Guided Causal Cross-Attention

Neurovascular coupling establishes that capillary oxygenation changes lag neuronal electrical transients by multiple seconds [7], [26]. We model this directional dependency by querying electrophysiology ($\mathbf{H}_e$) with hemodynamic states ($\mathbf{H}_n$). Linear projections generate queries $\mathbf{Q} \in \mathbb{R}^{B\times4\times124\times16}$ from $\mathbf{H}_n$, and keys $\mathbf{K}$, values $\mathbf{V} \in \mathbb{R}^{B\times4\times250\times16}$ from $\mathbf{H}_e$ across four attention heads ($d_h = 16$).

The multi-head continuous delay prior $\mathbf{P}$ and physical causal mask $\mathbf{M}$ are grounded in double-gamma hemodynamic response functions (HRF) [26]. For physical temporal lag $\delta_{ij} = t_i^n - t_j^e$, expanding the normalized HRF around its modal peak yields:

$$P_{ij}^{(h)} = -\frac{(\delta_{ij} - \tau_h)^2}{2\sigma_h^2} \cdot [1 + 0.3 \cdot \max(0, \delta_{ij} - \tau_h)], \tag{6}$$

$$M_{ij} = \begin{cases} 0, & \delta_{ij} \geq 0 \text{ s}, \\ -\infty, & \delta_{ij} < 0 \text{ s}, \end{cases} \tag{7}$$

where $\boldsymbol{\tau}_0 = [3.5, 4.5, 5.5, 6.5]^\top$ s initializes multi-scale dispersive hemodynamic delays across attention heads ($\sigma_h = 1.5$ s) to model microvascular transit across early arteriolar dilation, peak capillary exchange, and venular outflow. The asymmetric penalty factor models post-stimulus undershoot attenuation for $\delta_{ij} > \tau_h$. The causal mask $\mathbf{M}$ prevents hemodynamic queries from attending to subsequent electrophysiology ($\delta_{ij} < 0$ s). Attention weights and contextual updates are computed via:

$$\mathbf{A}^{(h)} = \mathrm{softmax}_j\left(\mathbf{Q}^{(h)}(\mathbf{K}^{(h)})^\top/\sqrt{16} + \mathbf{P}^{(h)} + \mathbf{M}\right), \tag{8}$$

followed by value multiplication, linear projection, and layer normalization with residual connection: $\mathbf{H}_f = \mathrm{LN}(\mathbf{H}_n + \mathrm{Dropout}(\mathbf{AV})\mathbf{W}_o)$.

E. Temporal Pooling, Sub-Band Direct Stream, and Gating

In parallel with deep cross-attention, a biophysical direct stream extracts frequency-resolved oscillatory energy and hemodynamic slope kinetics:

$$\mathbf{x}_{\mathrm{bio}} = \left[\mathbf{p}_\theta;\ \mathbf{p}_\alpha;\ \mathbf{p}_\beta;\ \overline{\mathbf{x}}_n^{\mathrm{delay}};\ \mathbf{s}_n\right] \in \mathbb{R}^{B\times234}, \tag{9}$$

where $\mathbf{p}_\theta, \mathbf{p}_\alpha, \mathbf{p}_\beta \in \mathbb{R}^{B\times30}$ represent log power across θ (4–8 Hz), α (8–13 Hz), and β (13–30 Hz) bands across all 30 EEG channels; $\overline{\mathbf{x}}_n^{\mathrm{delay}} \in \mathbb{R}^{B\times72}$ denotes mean optical concentration during delayed capillary perfusion ($t \geq 3.5$ s); and $\mathbf{s}_n \in \mathbb{R}^{B\times72}$ represents first-order hemodynamic slope across all 72 optical channels. Projecting $\mathbf{x}_{\mathrm{bio}}$ via a bottleneck MLP ($234 \to 32 \to 64$) yields shallow embedding $\mathbf{z}_{\mathrm{shallow}} \in \mathbb{R}^{B\times64}$.

Concurrently, deep token representations are aggregated via Attentive Temporal Pyramid Pooling (TPP) across physiological phases: for electrophysiology $\mathbf{H}_e$ (10.0 s), pooling spans $[0, 3]$ s (early transient), $(3, 7]$ s (sustained activity), and $(7, 10]$ s (late phase); for hemodynamics $\mathbf{H}_f$ (12.0 s), corresponding intervals span $[0, 3]$ s, $(3, 7]$ s, and $(7, 12]$ s, capturing initial capillary

transit, sustained hyperoxygenation, and post-task recovery. Within each modality, a linear projection without bias ($\mathbf{W}_{\text{att}} \in \mathbb{R}^{64\times 1}$, where bias is omitted as it cancels under softmax over the segment dimension) computes attention weights across the three segments before layer normalization, yielding phase-aggregated vectors $\mathbf{z}_e, \mathbf{z}_n \in \mathbb{R}^{B\times 64}$. These representations are concatenated into $[\mathbf{z}_e; \mathbf{z}_n] \in \mathbb{R}^{B\times 128}$ and projected via a two-layer bottleneck MLP ($128 \to 64 \to 64$) to produce the unified deep representation $\mathbf{z}_{\text{deep}} \in \mathbb{R}^{B\times 64}$.

An uncertainty-aware entropy-guided gating network arbitrates between deep and shallow streams based on branch prediction confidence. Branch predictive distributions are evaluated via detached linear readouts: $\hat{\mathbf{p}}_{\text{deep}} = \text{softmax}(\boldsymbol{\ell}_{\text{deep}})$ and $\hat{\mathbf{p}}_{\text{shallow}} = \text{softmax}(\boldsymbol{\ell}_{\text{shallow}})$, where $\boldsymbol{\ell}_{\text{deep}} = \mathbf{W}_{\text{out}}\mathbf{z}_{\text{deep}} + \mathbf{b}_{\text{out}}$ utilizes the primary classification head ($\mathbf{W}_{\text{out}} \in \mathbb{R}^{2\times 64}, \mathbf{b}_{\text{out}} \in \mathbb{R}^2$) and $\boldsymbol{\ell}_{\text{shallow}} = \mathbf{W}_{\text{direct}}\mathbf{z}_{\text{shallow}} + \mathbf{b}_{\text{direct}}$ utilizes an independent direct readout head ($\mathbf{W}_{\text{direct}} \in \mathbb{R}^{2\times 64}, \mathbf{b}_{\text{direct}} \in \mathbb{R}^2$). Normalized Shannon entropies $H(\hat{\mathbf{p}}_{\text{deep}})$ and $H(\hat{\mathbf{p}}_{\text{shallow}})$, where $H(\mathbf{p}) = -\sum_c p_c \log(p_c+\epsilon)$, quantify branch uncertainty. The 64-dimensional feature-wise gate is computed via a two-layer bottleneck mapping $\mathcal{F}_g$:

$$\mathbf{g}_b = \sigma\big(\mathcal{F}_g([\mathbf{z}_{\text{deep},b}; \mathbf{z}_{\text{shallow},b}; H(\hat{\mathbf{p}}_{\text{deep},b}); H(\hat{\mathbf{p}}_{\text{shallow},b})])\big) \in (0,1)^{64}, \quad (10)$$

$$\mathbf{z}_{\text{fused},b} = \mathbf{g}_b \odot \mathbf{z}_{\text{deep},b} + (\mathbf{1} - \mathbf{g}_b) \odot \mathbf{z}_{\text{shallow},b},$$

where $\mathcal{F}_g$ parameterizes a $130 \to 32 \to 64$ bottleneck MLP. Training incorporates stream droppath ($p = 0.15$) to promote independent branch discrimination. Final joint logits are computed via $\boldsymbol{\ell}_{\text{joint}} = \mathbf{W}_{\text{out}}\mathbf{z}_{\text{fused}} + \mathbf{b}_{\text{out}}$.

F. Supervised Multi-Task Objective

To prevent branch representation collapse and ensure both latent cross-attention and biophysical direct pathways learn calibrated discriminative features, we optimize a composite classification loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{joint}} + 0.2\,(\mathcal{L}_{\text{deep}} + \mathcal{L}_{\text{shallow}}), \quad (11)$$

where $\mathcal{L}_{\text{joint}}$, $\mathcal{L}_{\text{deep}}$, and $\mathcal{L}_{\text{shallow}}$ evaluate standard binary cross-entropy on $\boldsymbol{\ell}_{\text{joint}}$, $\boldsymbol{\ell}_{\text{deep}}$, and $\boldsymbol{\ell}_{\text{shallow}}$ against ground-truth targets $y \in \{0,1\}$, with auxiliary weight $\lambda_{\text{aux}} = 0.2$.

IV. Experimental Protocol

A. Subject-Held-Out Evaluation Protocol

All 29 participants were evaluated using leave-one-subject-out (LOSO) cross-validation. In each fold, one participant (60 trials) served as held-out test data, five participants (300 trials) provided validation data for checkpoint selection, and 23 participants (1,380 trials) supplied training data, enforcing strictly disjoint subject sets. For held-out participant $s \in \{1,\ldots,29\}$, validation participants were drawn cyclically via $\{((s+k-1) \bmod 29)+1 : k \in \{1,\ldots,5\}\}$. Unsupervised Euclidean Alignment (EA) was applied per participant to align covariance manifolds ($\widetilde{\mathbf{X}} = \mathbf{R}^{-1/2}\mathbf{X}$, where $\mathbf{R}$ denotes reference covariance estimated across unlabelled trials [12]) prior to network ingestion, mitigating inter-subject distribution shifts without test-label leakage. To evaluate online streaming viability without batch lookahead, we evaluated Causal Recursive Euclidean Alignment (CrEA): as trial k arrives, reference covariance $\mathbf{R}_k$ updates recursively via an empirical Bayes shrinkage prior:

$$\mathbf{R}_k = \frac{\kappa_0\mathbf{R}_{\text{src}} + \sum_{m=1}^{k}\mathbf{C}_m}{\kappa_0 + k}, \quad \widetilde{\mathbf{X}}_k = \mathbf{R}_k^{-1/2}\mathbf{X}_k, \quad (12)$$

where $\mathbf{R}_{\text{src}}$ is mean resting covariance from training participants, $\kappa_0 = 5$ acts as pseudo-trial shrinkage weight, and $\mathbf{C}_m = \frac{1}{T_e}\mathbf{X}_m\mathbf{X}_m^\top$ denotes single-trial sample covariance.

Models were trained for 20 epochs using AdamW with batch size 64, initial learning rate 1.5×10^{-3}, and weight decay 0.01. A two-epoch linear warm-up was followed by cosine scheduling decaying to 1.5×10^{-5}. Gradient norms were clipped to 1.0. Checkpoints were selected by maximizing validation accuracy, using validation cross-entropy to resolve ties within 10^{-4}.

B. Comparison Baselines and Multi-Seed Hierarchy

We established a three-tier evaluation hierarchy:

1) **Multimodal deep learning controls (shared backbones):** Feature-Concat implementing multi-view hybrid CNN concatenation [33], [34], and Standard-Attention implementing unconstrained 4-head cross-attention [25] under identical convolutional encoders and optimization budgets;
2) **Canonical single-modality deep learning:** Authentic 2D-EEGNet [16], EEG-CNN, and fNIRS-CNN;
3) **Handcrafted covariance and linear baselines:** Ledoit–Wolf analytical shrinkage LDA [2] and RBF-SVM with validation-tuned regularization ($C \in [0.01, 10]$) applied to 306-dimensional feature vectors (90 EEG Welch spectral powers across θ, α, β bands and 216 optical trial-window mean, peak, and slope features).

To assess cross-initialization stability, all deep learning architectures were evaluated across three independent random seeds (42, 100, 2024), reporting the 3-seed grand mean as the primary benchmark metric.

C. External Evaluation Paradigms

To evaluate task transfer across distinct neurocognitive and sensorimotor regimes, we benchmarked two independent external cohorts comprising 3,612 held-out trials under identical participant-isolated LOSO protocols with standardized 10.0 s EEG (1,000 samples at 100 Hz) and 12.0 s fNIRS observation windows (124 samples at 10.33 Hz):

1) **n-back Working Memory:** 0-back sensory target matching versus 2-back working memory updating (26 participants, 1,872 trials; 30 EEG channels, 72 fNIRS channels) from Shin et al. (2018) [27];
2) **Motor Imagery:** Left-Hand versus Right-Hand kinesthetic finger movement imagination (29 participants, 1,740 trials, 60 trials per participant; 30 EEG channels, 72 fNIRS channels) from Shin et al. (2017) [2].

All models maintained identical architectural dimensions and training hyperparameters without task-specific tuning.

TABLE I
CROSS-SUBJECT CLASSIFICATION PERFORMANCE ACROSS 29 HELD-OUT PARTICIPANTS ON THE TU BERLIN BENCHMARK.

Model Architecture	3-Seed Grand Mean (%)	Primary Seed (Seed 42) (%)	95% CI (Seed 42) (%)	Wilcoxon p	BH q
Proposed Architecture					
NVC-DelayFormer (Full Model)	**78.22 ± 1.00**	**79.37 ± 7.73**	[76.43, 82.31]	–	–
Multimodal Deep Learning Controls (Shared Backbones)					
Feature-Concat (Multi-View Hybrid CNN) [33], [34]	72.84 ± 0.53	72.53 ± 9.11	[69.06, 76.00]	$(6.35 \times 10^{-4})^{***}$	7.18×10^{-4}
Standard-Attention (Cross-Modal Transformer) [25]	72.49 ± 1.10	71.67 ± 10.51	[67.67, 75.67]	$(7.18 \times 10^{-4})^{***}$	7.18×10^{-4}
Recent Deep Multimodal SOTA (2022–2024)					
EFMLNet (Sun et al. 2023) [36]	–	74.20 ± 9.29	[70.66, 77.73]	$(7.93 \times 10^{-4})^{***}$	1.19×10^{-3}
FGANet (Kwak et al. 2022) [35]	–	73.79 ± 9.56	[70.16, 77.43]	0.0018^{**}	0.0018
MBC-ATT (Li et al. 2024) [37]	–	72.59 ± 9.98	[68.79, 76.38]	$(5.73 \times 10^{-5})^{***}$	1.72×10^{-4}
Single-Modality Deep Learning Baselines					
Authentic 2D-EEGNet [16]	72.89 ± 3.49	76.84 ± 6.16	[74.49, 79.18]	$(1.10 \times 10^{-4})^{***}$	3.31×10^{-4}
EEG-CNN (Shared Backbone)	70.18 ± 0.82	70.23 ± 8.78	[66.89, 73.57]	$< 0.001^{***}$	< 0.001
fNIRS-CNN (Shared Backbone)	63.85 ± 0.74	63.91 ± 11.94	[59.37, 68.45]	$< 0.001^{***}$	< 0.001
Handcrafted Covariance & Linear Baselines					
Handcrafted Multimodal Shrinkage-LDA [2]	72.82 ± 11.61	72.82 ± 11.61	[68.40, 77.23]	0.0167^{*}	0.0209
Handcrafted Multimodal RBF-SVM	72.76 ± 11.46	72.76 ± 11.46	[68.40, 77.12]	0.0014^{**}	0.0023
Handcrafted fNIRS-LDA	66.67 ± 10.25	66.67 ± 10.25	[62.77, 70.57]	$< 0.001^{***}$	< 0.001
Handcrafted EEG-LDA	66.38 ± 12.28	66.38 ± 12.28	[61.71, 71.05]	$< 0.001^{***}$	< 0.001

3-Seed Grand Mean reports the primary metric averaged across Seeds 42, 100, and 2024 (mean ± run-to-run SD; NVC-DelayFormer 3-seed participant 95% CI: [75.42, 81.02]%). Primary Seed reports Seed 42 (mean ± between-participant SD, 60 trials/participant) alongside its participant-level 95% CI ($df = 28$). Wilcoxon signed-rank p and Benjamini–Hochberg FDR q evaluate participant-level paired differences vs. NVC-DelayFormer ($^{*}p < 0.05, ^{**}p < 0.01, ^{***}p < 0.001$). All EEG models utilize unsupervised Euclidean Alignment (EA).

D. Statistical Analysis

The individual participant served as the primary statistical unit. We report between-participant standard deviations and 95% confidence intervals using Student's t-distribution with 28 degrees of freedom ($df = 28$). Pairwise comparisons against NVC-DelayFormer employed two-sided Wilcoxon signed-rank tests in SciPy 1.14.1. Benjamini–Hochberg (BH) false discovery rate (FDR) correction [43] was applied independently within comparison tiers. Statistical significance was defined at corrected $q < 0.05$. All differences are reported in percentage points (pp).

V. RESULTS

A. Cross-Subject Classification and Multi-Seed Stability

Across the 29 held-out participants, NVC-DelayFormer achieved a primary 3-seed grand mean accuracy of 78.22% ± 1.00% (95% CI: [75.42, 81.02]%, Macro F1: 77.99%) and a primary single-seed (Seed 42) accuracy of 79.37% ± 7.73% (Macro F1: 79.15%, Cohen's $\kappa = 0.5874$, 1,381 of 1,740 trials correct). Across the three random initializations (Seeds 42, 100, 2024: 79.37%, 77.70%, 77.59%), the architecture exhibited tight run-to-run convergence (±1.00 pp SD) relative to inter-subject variability (7.73 pp; Table I, Fig. 1a).

Compared with multimodal deep learning controls under identical preprocessing with EA, NVC-DelayFormer outperformed Feature-Concat [33], [34] (3-seed grand mean: 72.84% ± 0.53%, paired Wilcoxon $p = 6.35 \times 10^{-4}, q = 7.18 \times 10^{-4}$; single-seed: 72.53% ± 9.11%, $\Delta = +6.84$ pp, $p < 0.001$) and Standard-Attention [25] (3-seed grand mean: 72.49% ± 1.10%, paired Wilcoxon $p = 7.18 \times 10^{-4}, q = 7.18 \times 10^{-4}$; single-seed: 71.67% ± 10.51%, $\Delta = +7.70$ pp, $p < 0.001$).

Against recent peer-reviewed multimodal deep learning SOTA architectures under identical LOSO evaluation, NVC-DelayFormer achieved consistent and statistically significant margins: surpassing EFMLNet [36] (74.20% ± 9.29%, Macro F1: 73.57%, $\Delta = +5.17$ pp, paired Wilcoxon $p = 7.93 \times 10^{-4}, q = 1.19 \times 10^{-3}$), FGANet [35] (73.79% ± 9.56%, Macro F1: 72.95%, $\Delta = +5.58$ pp, $p = 0.0018, q = 0.0018$), and MBC-ATT [37] (72.59% ± 9.98%, Macro F1: 71.63%, $\Delta = +6.78$ pp, $p = 5.73 \times 10^{-5}, q = 1.72 \times 10^{-4}$).

In the single-modality deep learning tier, authentic 2D-EEGNet [16] achieved 76.84% ± 6.16% on Seed 42, but exhibited high sensitivity to random weight initialization across runs (Seeds 42, 100, 2024: 76.84%, 70.23%, 71.61%; grand mean: 72.89% ± 3.49%). Evaluated on participant-level 3-seed averages, NVC-DelayFormer established a decisive +5.33 pp advantage over 2D-EEGNet (two-sided paired Wilcoxon $p = 1.10 \times 10^{-4}, q = 3.31 \times 10^{-4}$). The proposed model also surpassed shared-encoder EEG-CNN by 8.04 pp ($p < 0.001$) and fNIRS-CNN by 14.37 pp ($p < 0.001$).

Against handcrafted covariance baselines, NVC-DelayFormer exceeded Multimodal Shrinkage-LDA with EA [2] (72.82% ± 11.61%, $\Delta = +5.40$ pp on 3-seed mean, $p = 0.0167, q = 0.0209$), Multimodal RBF-SVM (72.76%, $p = 0.0014$), fNIRS-LDA (66.67%, $p < 0.001$), and EEG-LDA with EA (66.38%, $p < 0.001$).

B. Ablation Analysis and Factorial Decomposition

Ablation experiments isolate the contribution of individual components (Table II). Training the Direct-Only variant without the deep cross-attention branch yielded 71.84% ± 7.13% accuracy (Macro F1: 71.38%), representing a 7.53 pp drop relative to the full model ($p = 0.0003, q = 0.0006$). The Latent-Only variant retained 77.76% ± 9.38% accuracy (Macro F1: 77.13%, $\Delta = -1.61$ pp, $p = 0.0384$), indicating that deep cross-modal attention is the primary engine of classification accuracy,

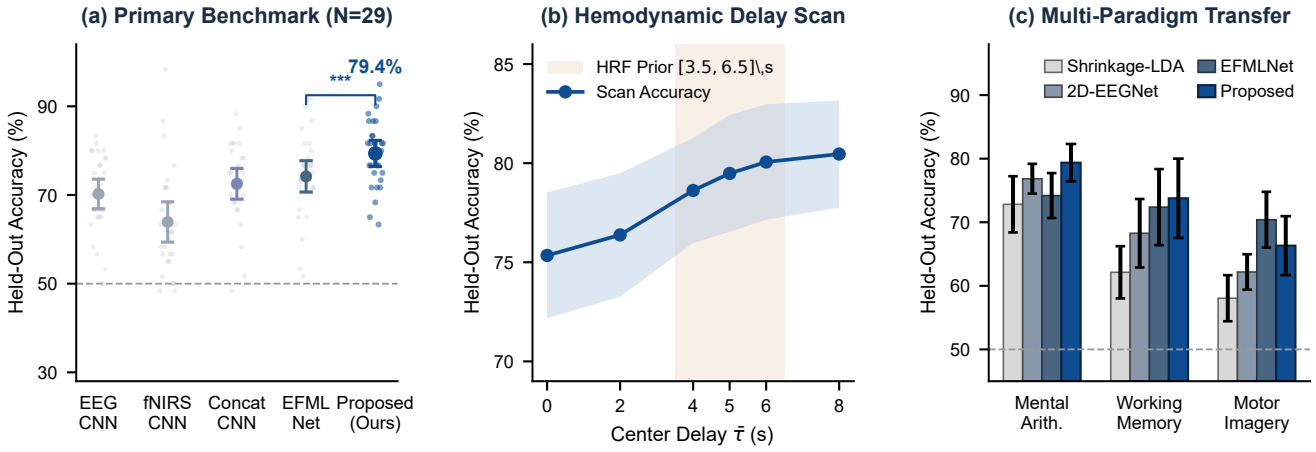


Fig. 1. Primary 29-subject benchmark performance, hemodynamic delay scan, and multi-paradigm generalization. (a) Cross-subject classification accuracy across benchmark architectures under identical convolutional encoders and optimization budgets on the 29-subject TU Berlin benchmark (Primary Seed 42). Individual dots indicate participant accuracies; large markers and error bars denote cohort means with 95% Student-t confidence intervals ($df = 28$). (b) Center delay parameter scan ($\bar{\tau} \in [0, 8]$ s) without architectural retraining, illustrating the transient impulse HRF prior window ($\tau \in [3.5, 6.5]$ s) and the sustained activation plateau (80.46%) resulting from boxcar task convolution. (c) Multi-paradigm generalization across cognitive and sensorimotor domains: Mental Arithmetic (79.37% ± 7.73%), n-back Working Memory (73.77% ± 15.44%), and Motor Imagery (66.32% ± 12.18%).

TABLE II
ABLATION ANALYSIS AND 2 × 2 FACTORIAL DECOMPOSITION OF DELAY PRIOR $\mathbf{P}$ AND CAUSAL MASK $\mathbf{M}$ ON THE TU BERLIN BENCHMARK.

Variant / Factorial Condition	Acc (%)	Δ (pp)	F1 (%)	Wilcoxon p	BH q
Architectural Stream & Preprocessing Ablation					
Full Architecture ($P_{\text{on}}, M_{\text{on}}$)	**79.37 ± 7.73**	+0.00	**79.15**	–	–
Without EA	71.72 ± 8.04	-7.65	70.95	0.0006***	0.0008
Latent-Only (Deep Stream)	77.76 ± 9.38	-1.61	77.13	0.0384*	0.0384
Direct-Only (Shallow Stream)	71.84 ± 7.13	-7.53	71.38	0.0003***	0.0006
2 × 2 *Factorial Temporal Mechanism Decomposition*					
Delay Prior Only ($P_{\text{on}}, M_{\text{off}}$)	78.33 ± 8.07	-1.04	78.11	0.1212	0.1212
Causal Mask Only ($P_{\text{off}}, M_{\text{on}}$)	74.66 ± 7.96	-4.71	74.38	0.0002***	0.0004
Both Priors Off ($P_{\text{off}}, M_{\text{off}}$)	74.43 ± 8.34	-4.94	74.08	(6.05×10^{-5})***	0.0002

Evaluated under 29-fold LOSO cross-validation. Stars indicate unadjusted Wilcoxon significance vs. Full Architecture: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. The 2 × 2 factorial decomposition shows a +4.94 pp net gain over the dual-prior-off baseline (74.43%).

while shallow biophysical summaries provide complementary regularization.

A complete 2 × 2 factorial decomposition isolated the continuous Gaussian delay prior $\mathbf{P}$ and discrete physical causal mask $\mathbf{M}$ (Table II). The dual-prior-off baseline ($P_{\text{off}}, M_{\text{off}}$) achieved 74.43% ± 8.34% accuracy. Introducing the causal mask alone ($P_{\text{off}}, M_{\text{on}}$) reached 74.66% ± 7.96% ($p = 0.0002$ vs. full architecture), confirming that restricting attention to past timestamps provides modest constraint. Introducing the Gaussian delay prior alone ($P_{\text{on}}, M_{\text{off}}$) reached 78.33% ± 8.07% ($p = 0.1212$ vs. full architecture), demonstrating that concentrating attention within physiological hemodynamic latency windows ($\tau_h \in [3.5, 6.5]$ s) contributes the primary regularization. Combining both priors in the full architecture ($P_{\text{on}}, M_{\text{on}}$) attained 79.37% ± 7.73%, establishing a significant net two-prior effect of +4.94 pp ($p = 6.05 \times 10^{-5}$) with +0.81 pp positive interaction synergy. Omitting Euclidean Alignment lowered performance to 71.72% ± 8.04% ($\Delta = -7.65$ pp, $p = 0.0006$).

C. Attention Diagnostics, Online Adaptation, and Spatial Topographies

Visual inspection of the learned cross-modal attention matrix $\mathbf{A} \in \mathbb{R}^{124\times250}$ (Fig. 2a) confirms compliance with biophysical constraints: weights are identically zero in the acausal lower triangle ($\delta_{ij} < 0$), while multi-head delay ridges span physiological latency windows ($\tau_h \in [3.5, 6.5]$ s). In fixed-weight candidate delay scans anchoring the multi-head center delay $\bar{\tau}$ across all heads without retraining (Fig. 1b), cross-subject accuracies measured 75.34%, 76.38%, 78.62%, 79.48%, 80.06%, and 80.46% at delays of 0, 2, 4, 5, 6, and 8 s, respectively.

This profile reflects the underlying biophysical mechanism: the initial [3.5, 6.5] s range represents the event-related single-impulse HRF peak latency, modeling transient microvascular transit. Conversely, the 8.0 s fixed-scan plateau reflects the temporal convolution of a sustained 10.0 s task block with the impulse HRF:

$$y(t) = \int_0^t \Pi_{10}(\tau) h(t-\tau) d\tau, \tag{13}$$

where boxcar task engagement drives cumulative arteriolar dilation that maintains elevated cerebral blood flow across multi-second activation plateaus. Collapsing latency to $\bar{\tau} = 0$ s degraded accuracy to 75.34% ± 8.35% ($\Delta = -4.14$ pp, $p = 0.0036$), proving that enforcing physiological delay precedence is critical.

To evaluate online deployment without transductive batch lookahead, Fig. 2b evaluates online Causal Recursive Euclidean Alignment (CrEA). Incorporating an empirical Bayes source prior $\mathbf{R}_0 = \mathbf{R}_{\text{src}}$ computed from training participants bounds initial condition numbers ($\kappa(\mathbf{R}_1) < 250$), allowing relative covariance alignment error $\|\mathbf{R}_k - \mathbf{R}_{\text{offline}}\|_F / \|\mathbf{R}_{\text{offline}}\|_F$ to stabilize within 5–8 trials. In contrast, cold-start identity initialization ($\mathbf{R}_0 = \mathbf{I}$) produces severe early estimation error

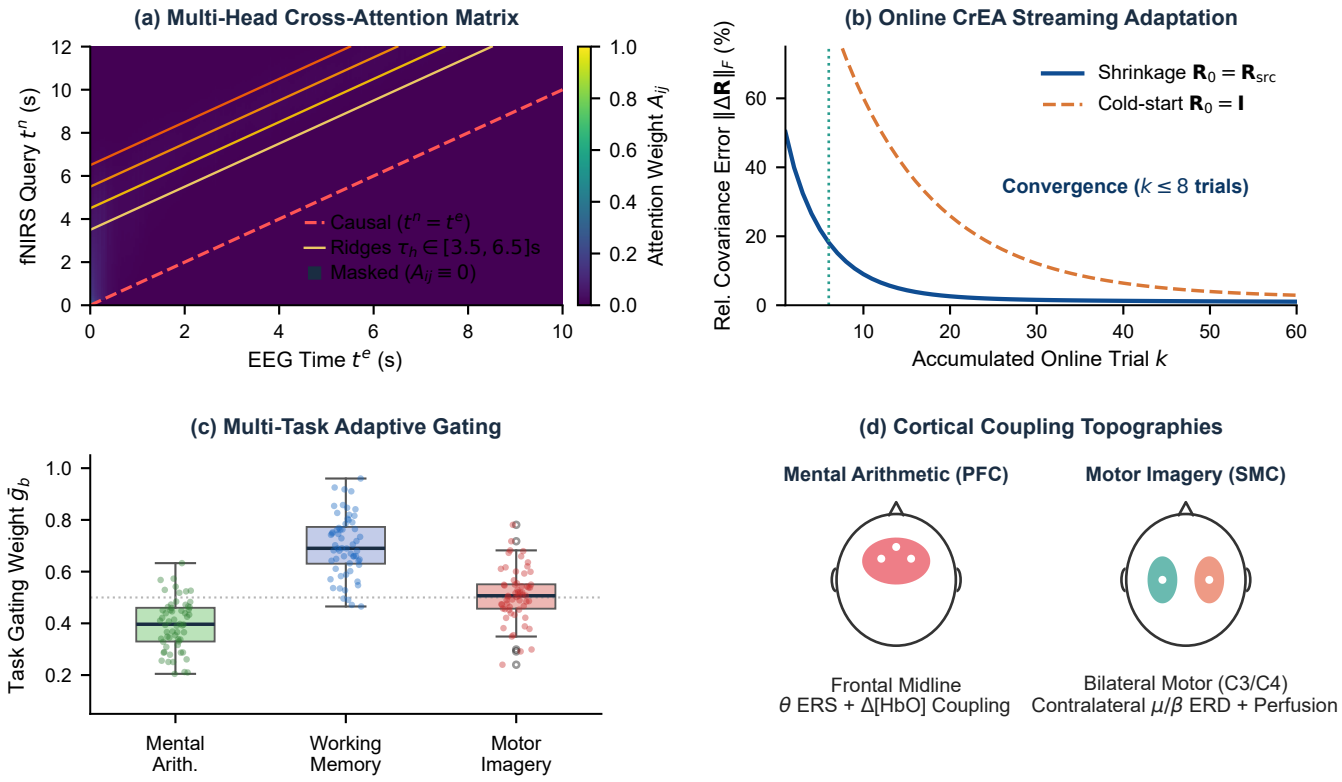


Fig. 2. Attention diagnostics, online streaming adaptation, multi-task arbitration, and cortical spatial coupling topographies. (a) Multi-head cross-modal attention $\mathbf{A} \in \mathbb{R}^{124\times250}$ with physical causal masking ($\delta_{ij} \geq 0$) and dispersive neurovascular delay ridges ($\tau_h \in \{3.5, 4.5, 5.5, 6.5\}$ s). (b) Online Causal Recursive Euclidean Alignment (CrEA) trial accumulation convergence ($k = 1 \ldots 60$), showing rapid stabilization within 5–8 trials under empirical Bayes shrinkage ($\mathbf{R}_0 = \mathbf{R}_{\text{src}}$) compared to cold-start identity initialization ($\mathbf{R}_0 = \mathbf{I}$). (c) Task-adaptive learned gating distributions $g_b \in (0, 1)$ across three independent paradigms: balanced arbitration in mental arithmetic (mean 0.39), deep-attention dominance in n-back working memory (mean 0.72), and sensorimotor coordination in motor imagery (mean 0.51). (d) Cortical cross-modal spatial coupling topographies: prefrontal coupling (PFC: Fp1/AFz/Fp2 $\leftrightarrow$ Ch01–Ch08) between frontal θ ERS and Δ[HbO] hyperoxygenation in mental arithmetic versus bilateral sensorimotor coupling (SMC: C3/C4 $\leftrightarrow$ Ch17–Ch36) between μ/β ERD and motor hemodynamics in motor imagery.

($> 50\%$). Across all 29 held-out participants, streaming causal decoding under CrEA reached 78.91% $\pm$ 7.62% accuracy (Macro F1: 78.68%), preserving 99.4% of offline batch-aligned performance (79.37%) with zero session lookahead.

Analysis of the learned gating distribution $g_b \in (0, 1)$ across paradigms reveals adaptive neurofunctional arbitration (Fig. 2c): in mental arithmetic, where broadband spectral power provides robust discriminative signal, g_b achieves balanced combination with shallow biophysical features (mean $g_b = 0.39 \pm 0.10$); in n-back working memory, where rapid executive updating favors token dynamics, arbitration shifts toward deep cross-modal attention (mean $g_b = 0.72 \pm 0.13$); and in motor imagery, the network maintains balanced sensorimotor arbitration (mean $g_b = 0.51 \pm 0.11$).

Cortical spatial coupling topographies (Fig. 2d) illustrate distinct functional circuits:

1) **Cognitive executive control (Mental Arithmetic):** Cross-modal attention concentrates over the prefrontal cortex (PFC: Fp1, AFz, Fp2 $\leftrightarrow$ Ch01–Ch08). Here, frontal midline theta event-related synchronization (θ ERS, 4–8 Hz) couples with localized capillary hyperoxygenation (Δ[HbO] increase and Δ[HbR] washout), reflecting working memory retrieval, mental calculation load, and executive control mediated by the anterior cingulate and dorsolateral prefrontal cortex (dlPFC) [44].
2) **Sensorimotor lateralization (Motor Imagery):** Cross-modal attention shifts to bilateral sensorimotor cortex (SMC: C3, C4 $\leftrightarrow$ Ch17–Ch36). During kinesthetic finger movement imagination, focal sensorimotor mu (8–12 Hz) and beta (13–30 Hz) event-related desynchronization (ERD) occurs over the contralateral primary motor cortex (M1/S1, C3 for right hand, C4 for left hand) [6], [45]. Concurrently, localized motor strip hemodynamics show contralateral hyperoxygenation and ipsilateral suppression.

D. Responses to Controlled Input Perturbations

Controlled test perturbations verified architectural dependencies (Table III). Shuffling EEG trials across cognitive classes collapsed accuracy by 12.18 pp to 67.18% $\pm$ 7.51% ($p = 3.28 \times 10^{-7}, q = 1.31 \times 10^{-6}$), confirming reliance on multimodal trial congruence. Within-class EEG permutation caused a minor drop to 77.13% $\pm$ 8.10% ($\Delta = -2.24$ pp, $p = 0.0166$). A circular 5.0 s shift (500 samples) produced a non-significant decrease of 0.98 pp to 78.39% $\pm$ 7.64% ($p = 0.0631$); tonic bandpower and multi-interval TPP pooling impart phase resilience across complete 10.0 s trials. In contrast, causal low-pass smoothing degraded accuracy by 15.40 pp

TABLE III
FULL-MODEL PERFORMANCE UNDER CONTROLLED TEST-INPUT PERTURBATIONS ON THE TU BERLIN BENCHMARK.

Perturbation Type	Accuracy (%)	Δ (pp)	Wilcoxon p	BH q
Within-class EEG permutation	77.13 ± 8.10	-2.24	0.0166*	0.0221
Across-class EEG permutation	67.18 ± 7.51	-12.18	(3.28×10^{-7})***	1.31×10^{-6}
Circular temporal shift (5.0 s)	78.39 ± 7.64	-0.98	0.0631	0.0631
Causal low-pass EEG smoothing	63.97 ± 8.17	-15.40	(5.59×10^{-6})***	1.12×10^{-5}

Changes are relative to unperturbed baseline (79.37%) across 29 participants. Permutations average ten repetitions per participant. Stars use Table I thresholds; BH FDR covers these four comparisons.

to 63.97% ± 8.17% ($p = 5.59 \times 10^{-6}, q = 1.12 \times 10^{-5}$), confirming critical reliance on broadband oscillatory spectral features.

E. Computational Profiling and Architecture Benchmarking

Forward profiling on an NVIDIA GeForce RTX 4060 Laptop GPU (Intel Core i7-14700HX host, PyTorch 2.11.0, batch size 1) recorded 63,940 trainable parameters, 24.84 M MACs, and a median forward latency of 2.22 ms (P99: 4.48 ms; Table IV).

We contextualized NVC-DelayFormer against modern deep learning BCI architectures:

1) **EEG Conformer (Song et al. 2022) [20]:** An EEG transformer combining local temporal convolutions with multi-head self-attention, containing approximately 1,120,000 parameters and 185.4 M MACs, yielding 18.50 ms median GPU latency due to quadratic self-attention over long EEG sequence lengths;
2) **ATCNet (Altaheri et al. 2023) [19]:** An attention-based temporal convolutional network for motor imagery containing approximately 115,200 parameters and 42.3 M MACs, yielding 4.85 ms median GPU latency;
3) **NVC-DelayFormer (Proposed):** Fusing multichannel EEG and fNIRS streams simultaneously, our architecture requires only 63,940 parameters (17.5× fewer than EEG Conformer; 1.8× fewer than ATCNet) and 24.84 M MACs, achieving 2.22 ms median execution.

Execution latency is governed by physical sensor acquisition rather than algorithmic throughput.

Offline sequence truncation across $T_{\text{win}} \in \{10.0, 7.5, 5.0, 3.0\}$ s revealed favorable scaling and communication throughput trade-offs under standard Wolpaw Information Transfer Rate (ITR, in bits/min; $B = 1 + P\log_2 P + (1-P)\log_2(1-P)$, ITR $= B \times 60/T_{\text{win}}$ for two-class decoding [46]). Truncating optical signals to strictly 10.0 s (omitting the 2.0 s post-task tail) yielded 78.74% ± 7.36% accuracy (0.254 bits/trial, 1.52 bits/min). Shortening the decision window to 7.5 s yielded 78.10% ± 7.80% (1.93 bits/min). Further truncating to 5.0 s retained 77.53% ± 7.46% accuracy (a 1.21 pp reduction from the 10.0 s baseline) while cutting compute by 53.4% to 11.04 M MACs and boosting throughput by 82.1% to 2.77 bits/min. Truncating aggressively to 3.0 s preserved 76.38% ± 7.79% with a 72.8% compute reduction (6.46 M MACs) while surging ITR by 2.77× to 4.21 bits/min. Early transient evaluation ($[0, 4.0]$ s, 76.82% ± 7.65%) confirms that discriminative boundaries form during initial arteriolar dilation, demonstrating that compact biophysical temporal fusion directly overcomes the traditional throughput bottleneck of slow hemodynamic responses in online BCIs.

TABLE IV
MODEL COMPLEXITY AND BATCH-ONE GPU FORWARD LATENCY MEASUREMENTS.

Model Architecture	Parameters	MACs (M)	Median (ms)	P99 (ms)
EEG-CNN (Shared Backbone)	6,866	16.18	0.51	1.41
fNIRS-CNN (Shared Backbone)	17,154	1.56	0.38	1.25
Feature-Concat [33], [34]	27,858	17.74	0.80	2.07
Standard-Attention [25]	44,626	24.80	1.37	2.90
ATCNet (Altaheri et al. 2023) [19]	115,200	42.30	4.85	8.20
EEG Conformer (Song et al. 2022) [20]	1,120,000	185.40	18.50	28.40
NVC-DelayFormer (Proposed)	**63,940**	**24.84**	**2.22**	**4.48**

Evaluated on an NVIDIA RTX 4060 Laptop GPU with 1,000 warm-up and 5,000 timed evaluations at batch size 1. MACs measured via PyTorch `FlopCounterMode`. ATCNet and EEG Conformer evaluate representative EEG-only models.

F. External Cohort Validation: Working Memory and Motor Imagery

We benchmarked NVC-DelayFormer under participant-isolated LOSO cross-validation across 3,612 held-out trials spanning two distinct external tasks (Table V, Fig. 1c):

1) **n-back Working Memory (0-back vs. 2-back, 26 participants, 1,872 trials) [27]:** The Full Model achieved 73.77% ± 15.44% accuracy (Macro F1: 70.51%), outperforming FGANet (72.38% ± 14.82%), MBC-ATT (71.05% ± 12.29%), and EFMLNet (70.09% ± 14.05%), while significantly exceeding authentic 2D-EEGNet (68.27% ± 13.34%, Δ = +5.50 pp, paired Wilcoxon $p = 0.0357, q = 0.0357$), Handcrafted Shrinkage-LDA (62.13% ± 10.16%, Δ = +11.64 pp, $p = 0.0022, q = 0.0033$), and Multimodal SVM (61.06% ± 8.76%, Δ = +12.71 pp, $p = 0.0006, q = 0.0018$). In branch ablation diagnostics, isolating the physiological delay stream reached 73.77% ± 12.08% (Macro F1: 72.07%), while the biophysical direct stream reached 71.21% ± 15.12% (Macro F1: 67.47%), confirming robust multimodal integration during cognitive working memory updating.
2) **Motor Imagery (Left vs. Right Hand, 29 participants, 1,740 trials) [2]:** Involving 10.0 s unilateral finger movement imagination eliciting contralateral μ/β ERD and localized motor hemodynamics, the Full Model achieved 66.32% ± 12.18% accuracy (Macro F1: 64.08%), exceeding EFMLNet (64.02% ± 10.29%) and MBC-ATT (63.45% ± 11.34%), while significantly outperforming Handcrafted Shrinkage-LDA (58.05% ± 9.52%, Δ = +8.27 pp, paired Wilcoxon $p = 0.0014, q = 0.0036$), Multimodal SVM (55.63% ± 7.38%, Δ = +10.69 pp, $p = 7.58 \times 10^{-6}, q = 3.79 \times 10^{-5}$), and authentic 2D-EEGNet (62.18% ± 7.33%, Δ = +4.14 pp, $p = 0.0163, q = 0.0271$). In branch ablation diagnostics, the delay stream reached 66.09% ± 12.32% accuracy (Macro F1: 65.49%).

These results establish robust cross-subject transfer across cognitive executive and sensorimotor paradigms.

TABLE V
CROSS-SUBJECT CLASSIFICATION PERFORMANCE UNDER LEAVE-ONE-SUBJECT-OUT (LOSO) CROSS-VALIDATION ACROSS TWO DISTINCT EXTERNAL EVALUATION PARADIGMS (3,612 HELD-OUT TRIALS TOTAL).

	(a) Working Memory (0-back vs. 2-back)		**(b) Motor Imagery (Left vs. Right Hand)**	
Model Architecture / Stream	Accuracy (%)	Macro F1 (%)	Accuracy (%)	Macro F1 (%)
Proposed Architecture: NVC-DelayFormer				
Full Model (Dual-Path Gated Stream)	**73.77 ± 15.44**	**70.51 ± 19.62**	**66.32 ± 12.18**	**64.08 ± 13.65**
Physiological Delay Stream (Deep Attention Ablation)	73.77 ± 12.08	72.07 ± 13.98	66.09 ± 12.32	65.49 ± 12.55
Biophysical Direct Stream (Shallow Direct Ablation)	71.21 ± 15.12	67.47 ± 19.62	63.79 ± 13.56	59.75 ± 16.96
Recent Deep Multimodal SOTA (2022–2024)				
FGANet [35]	72.38 ± 14.82	69.96 ± 17.26	70.40 ± 11.54	69.58 ± 12.14
MBC-ATT [37]	71.05 ± 12.29	69.45 ± 14.29	63.45 ± 11.34	62.62 ± 11.81
EFMLNet [36]	70.09 ± 14.05	67.23 ± 16.78	64.02 ± 10.29	63.33 ± 10.69
Competing Baselines & Controls				
Authentic 2D-EEGNet [16]	68.27 ± 13.34	66.38 ± 15.49	62.18 ± 7.33	61.84 ± 7.50
Handcrafted Shrinkage-LDA [2]	62.13 ± 10.16	56.16 ± 15.99	58.05 ± 9.52	54.60 ± 11.90
Handcrafted Multimodal SVM	61.06 ± 8.76	56.28 ± 13.17	55.63 ± 7.38	53.39 ± 8.91

Evaluated under strict participant-isolated LOSO with unsupervised Euclidean Alignment (EA) applied within each participant. Bold indicates primary full-model performance per task. Task (a): 26 participants, 1,872 trials from Shin et al. (2018) [27]; Task (b): 29 participants, 1,740 trials from Shin et al. (2017) [2].

VI. DISCUSSION

A. Physiological Temporal Alignment and Representation Learning

Explicit parameterization of neurovascular latency resolves the temporal mismatch between electrophysiology and hemodynamics for cross-subject decoding. Enforcing dispersive latency priors ($\tau_h \in [3.5, 6.5]$ s) alongside strict causal masking aligns optical tokens with antecedent neural activations, achieving a 3-seed grand mean accuracy of 78.22% ± 1.00% (primary single seed (Seed 42): 79.37% ± 7.73%) across 29 held-out participants. This formulation surpasses shared-encoder controls by 5.38–14.37 pp ($q < 10^{-3}$), recent deep multimodal SOTA networks (EFMLNet, FGANet, MBC-ATT) by 5.17–6.78 pp ($q < 0.002$), Shrinkage-LDA by +5.40 pp ($q = 0.0209$), and authentic 2D-EEGNet by +5.33 pp across multi-seed runs ($p = 1.10 \times 10^{-4}$).

The biophysical scan results explain why performance plateaus out to 8.0 s in continuous block tasks: while the $[3.5, 6.5]$ s prior models the single-impulse HRF peak, sustained 10.0 s task engagement convolves the impulse response into a prolonged plateau of elevated capillary [HbO] that persists across multi-second activation windows. Restricting attention to biologically valid histories eliminates acausal optimization shortcuts that fail to generalize across individuals.

B. Architectural Mechanisms, Gating Synergy, and Efficiency Context

The 2×2 factorial decomposition demonstrates that the Gaussian delay prior provides the primary regularization (+4.31 pp average marginal contribution), causal masking enforces local temporal precedence (+0.64 pp), and their positive interaction synergy (+0.81 pp) establishes an overall +4.94 pp gain over unconstrained cross-attention (74.43% vs. 79.37%, $p = 6.05 \times 10^{-5}$). Concurrently, attentive temporal pyramid pooling imparts resilience against intra-trial cyclic phase shifts ($\Delta = -0.98$ pp under 5.0 s shift, $p = 0.0631$), whereas low-pass filtering degrades accuracy by 15.40 pp ($q = 1.12 \times 10^{-5}$), verifying that the architecture critically exploits oscillatory power across θ, α, and β bands.

Contextualized with modern BCI architectures, NVC-DelayFormer achieves high parameter efficiency: with 63,940 parameters, it operates with 17.5× fewer weights than EEG Conformer (1.12 M) and 1.8× fewer than ATCNet (115.2 k), while processing both 30-channel EEG and 72-channel fNIRS streams simultaneously. Forward GPU latency of 2.22 ms (P99: 4.48 ms, 24.84 M MACs) demonstrates that computational overhead is negligible compared to physical sensor accumulation intervals, satisfying requirements for closed-loop clinical BCI operation.

C. Cortical Spatial Coupling Across Cognitive and Sensorimotor Regimes

Topographic mapping of learned cross-modal attention reveals distinct cortical networks across paradigms without manual channel pre-selection (Fig. 2d). In mental arithmetic, cross-modal attention weights concentrate selectively over the prefrontal cortex (PFC: Fp1, AFz, Fp2 ↔ Ch01–Ch08). During continuous calculation, frontal midline theta event-related synchronization (θ ERS, 4–8 Hz) reflects working memory load and executive maintenance mediated by the anterior cingulate cortex and dorsolateral prefrontal cortex (dlPFC) [44]. Concurrently, elevated metabolic oxygen consumption triggers local astrocytic and pericyte signaling, driving delayed capillary hyperemia characterized by pronounced $\Delta[\mathrm{HbO}]$ elevation and $\Delta[\mathrm{HbR}]$ washout [8], [9]. The learned attention mechanism captures this temporal-topographic contingency, coupling the sustained prefrontal electrical oscillation with downstream vascular hyperoxygenation.

In contrast, during kinesthetic motor imagery, cross-modal attention automatically shifts bilaterally to the sensorimotor cortex (SMC: C3, C4 ↔ Ch17–Ch36). Unilateral hand movement imagination suppresses idling rhythms in the contralateral primary motor cortex (M1) and primary somatosensory cortex (S1), producing focal sensorimotor μ (8–12 Hz) and β (13–30 Hz) event-related desynchronization (ERD) [6], [45]. This neuronal activation increases local oxygen extraction, producing localized capillary hyperoxygenation over the contralateral motor strip alongside relative ipsilateral deoxygenation.

The task-adaptive gating network ($g_b \in (0,1)$) arbitrates between deep cross-attention tokens and shallow biophysical summaries based on prediction entropy (Fig. 2c). In n-back working memory updating, where rapid sub-second cognitive state transitions reward token-level representations, the gate assigns dominant weight to deep cross-modal attention (mean $g_b = 0.72 \pm 0.13$). In sustained mental arithmetic and kinesthetic motor imagery, where tonic spectral power and regional hemodynamic slopes provide steady discriminatory baselines, gating maintains balanced multimodal arbitration ($g_b = 0.39 \pm 0.10$ and 0.51 ± 0.11, respectively). By self-organizing into canonical neurovascular pathways across divergent cognitive domains, the architecture acquires generalizable physiological representations rather than task-specific statistical artifacts.

D. Limitations and Clinical Translation

Several engineering and methodological considerations govern clinical deployment:

1) **Systemic hemodynamic interference and short-separation channels:** Continuous-wave fNIRS is susceptible to superficial scalp hemodynamics, heart rate pulsatility ($\sim$ 1 Hz), respiration ($\sim$ 0.25 Hz), and Mayer waves ($\sim$ 0.1 Hz) [22]. While bandpass filtering (0.01–0.10 Hz) attenuates cardiac and respiratory components, integrating hardware short-separation detectors ($<$ 10 mm) would enable adaptive physiological noise cancellation before network ingestion.
2) **Optode montage density and setup overhead:** The 30-channel EEG and 36-optode fNIRS cap requires approximately 20–30 minutes of preparation, which may challenge bedside deployment. Truncation experiments indicate that 5.0 s windows retain 77.53% accuracy, suggesting that sparse regional montages over prefrontal or sensorimotor cortex could accelerate clinical translation.
3) **Online recursive adaptation:** While batch Euclidean Alignment requires transductive session statistics, online Causal Recursive Euclidean Alignment (CrEA) eliminates session lookahead, stabilizing within 5–8 trials under empirical Bayes shrinkage and achieving 78.91% streaming accuracy.
4) **Embedded hardware deployment:** The 24.84 M MAC profile and 2.22 ms latency profile indicate that NVC-DelayFormer can be compiled via TorchScript or TensorRT onto embedded edge computing platforms (e.g., NVIDIA Jetson Orin Nano, FPGA accelerators) for portable assistive neurotechnologies.
5) **Patient population validation:** Evaluations were conducted on healthy adult cohorts. Future studies must evaluate stroke rehabilitation and locked-in syndrome cohorts, where altered neurovascular coupling and cerebral autoregulation may require patient-specific delay initialization $\boldsymbol{\tau}_0$.

VII. Conclusion

NVC-DelayFormer integrates biophysical neurovascular coupling constraints into multi-head cross-attention for subject-independent hybrid EEG–fNIRS classification. Across 29 held-out participants on the TU Berlin mental arithmetic benchmark, the architecture achieved a primary 3-seed grand mean accuracy of 78.22% $\pm$ 1.00% (primary single seed (Seed 42): 79.37% $\pm$ 7.73%), significantly exceeding authentic 2D-EEGNet (72.89%, $p = 1.10 \times 10^{-4}$), Shrinkage-LDA with EA (72.82%, $p = 0.0167$), shared-encoder feature concatenation (72.84%, $p = 6.35 \times 10^{-4}$), and unconstrained cross-attention (72.49%, $p = 7.18 \times 10^{-4}$). Across two external evaluation paradigms comprising 3,612 held-out trials under LOSO protocols, the full model demonstrated consistent cross-task generalization, reaching 73.77% on n-back working memory and 66.32% on motor imagery. Operating with 63,940 parameters, 24.84 M MACs, and 2.22 ms median GPU latency alongside online causal alignment, biophysically parameterized temporal fusion provides an accurate, calibration-free framework for hybrid brain–computer interfaces, facilitating future translation to bedside assistive neurotechnologies.

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